Project 1 - Sorting Algorithms

CS 2430 | Connor | Summer 2026

Description

This project implements and compares four sorting algorithms in C++:

- Merge Sort
- Quick Sort
- Heap Sort
- Shaker Sort

Each algorithm tracks the number of comparisons made during sorting.
Results are displayed in a table showing min, max, and average comparisons
over 10 runs for array sizes n = 4, 6, and 8.

Language

C++ (C++17)

Build Instructions

From the `/project1` directory:

```
cmake -S . -B build
cmake --build build
```

Run Instructions

```
.\build\SortingAlgorithms.exe
```

File Structure

```
project1/
├── src/
│   ├── main.cpp                  - test driver and output
│   ├── algorithms/
│   │   ├── sorting.cpp           - merge, quick, heap, and shaker sort
│   │   └── generator.cpp         - random array generator
│   └── include/
│       ├── sorting.h             - sorting function declarations
│       └── generator.h           - generator function declaration
│
```

```
├── CMakeLists.txt                - build configuration
└── README.md                     - this file
```